

10 - The Plus Operator Boundary and Protected Incompleteness v0.1

Academic PDF rendering for the CCALC / C-calculus governed binding stack v0.1.

Document ID	10_THE_PLUS_OPERATOR_BOUNDARY_AND_PROTECTED_INCOMPLETENESS_v0_1.md
Version/status	normative draft / theorem-form profile / perimeter-closure interpretation
Stack layer	central operator-boundary / protected incompleteness / non-collapse theorem-form profile
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Reserved DOI	10.5281/zenodo.21205427
Notice	This DOI is reserved and becomes active after Zenodo publication.

Abstract / purpose

Documents 01 through 09 formalize the perimeter around $c = a + b$: This document defines the central boundary those layers protect: The purpose is not to turn $+$ into an ordinary executable operation. The purpose is to define why $+$ must remain a protected, non-collapsing boundary that cannot be reduced to a total substrate-computable function without destroying the meaning of: This profile therefore answers: ---

Non-claims boundary

- _ not legal advice
- _ not privacy-law certification
- _ not safety certification
- _ not deployment authorization
- _ not standards compliance certification
- _ not regulated approval
- _ not C-A1 ratification
- _ not personhood proof
- _ not consciousness proof
- _ not live substrate truth
- _ not proof of completeness

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10 - The Plus Operator Boundary and Protected Incompleteness v0.1

Artifact: 10_THE_PLUS_OPERATOR_BOUNDARY_AND_PROTECTED_INCOMPLETENESS_v0_1.md Package: CCALC_PLUS_OPERATOR_BOUNDARY_10_v0_1 Project: Self-Evo / Ester / $c = a + b$ Layer: central operator-boundary / protected incompleteness / non-collapse theorem-form profile Status: normative draft / theorem-form profile / perimeter-closure interpretation Created UTC: 2026-07-05 Author: Kotov Ivan Claim-force ceiling: C-A4 normative/theorem-form profile + C-A10 corpus-control profile; C-A7 only for explicit hash/custody records. This document does not ratify C-A1.

0. Purpose

Documents 01 through 09 formalize the perimeter around $c = a + b$:

```
01 algebra and governed operator family
02 binding operator and anchor review
03 transition semantics
04 continuity / equivalence / fork / replay / archive
05 self-evolution / bounded growth
06 runtime authority / multi-contour deployment
07 public evidence / redaction / release custody
08 interoperability / external review intake
09 deployment / regulated release boundary
```

This document defines the central boundary those layers protect:

+

The purpose is not to turn + into an ordinary executable operation.

The purpose is to define why + must remain a protected, non-collapsing boundary that cannot be reduced to a total substrate-computable function without destroying the meaning of:

$c = a + b$

This profile therefore answers:

```
What kind of object is +?
What must never be confused with +?
What does the surrounding corpus prove about +?
Why is the incompleteness of + a protected feature rather than a defect?
What checker consequences follow when any layer tries to simulate + through publication, review,
deployment, reproduction, memory, or model output?
```

0.1 Executive definition

+ is the governed binding boundary between an accountable anchor a and a technological substrate b.

It is not arithmetic addition.

It is not merge.

It is not an API call.

It is not a certificate by itself.

It is not public release, deployment approval, external review, reproduction pass, model confidence, memory continuity, or standards mapping.

Operationally:

$+_g(a,b) \rightarrow c$

means:

```
an accountable anchor and a bounded substrate are joined under declared governance,
without collapsing anchor into substrate,
without letting substrate emulate anchor,
and without letting evidence surfaces become authority surfaces.
```

The protected rule:

```
The perimeter around + may be formalized.
The center may not be reduced to b-computation.
```

A system may provide evidence that a binding event is admissible. It may not compute the fact of binding solely from inside b.

0.2 Mirror of the single-operator formalization bridge

The working input behind this document compares two opposite uses of a single operator.

One direction:

```
small primitive operator
+ grammar / construction rules
-> expressive completeness
```

The c direction:

```
complete governed perimeter
+ protected non-reducible center
-> non-collapse of anchor, substrate, authority, and continuity
```

The structural contrast is:

```
single operator as source of constructive completeness
versus
single operator as protected boundary against illegitimate completeness
```

This is not a rejection of formalization.

It is a sharper formalization target:

```
formalize everything that may be formalized around +;
prove, by the corpus invariants, that reducing + to b would violate those invariants.
```

The useful sentence:

```
+ is not what computes c.
+ is what prevents c from being reduced to b.
```

0.3 Relationship to 01 and 02

01 defines the typed governed algebra and the operator family:

```
+_g : Anchor × Substrate -> CState
```

02 isolates the governed binding operator as a partial operator over anchor, substrate, and governance profile.

This document does not replace those layers.

It clarifies the deepest boundary behind them:

```
The operator profile can define preconditions, certificates, review surfaces, envelopes, leases,
witnesses, and failure states.
It cannot allow b to synthesize the anchor-grounded binding event by itself.
```

Therefore +_g has two faces:

Formal perimeter

```
input contracts
anchor signature envelope
substrate manifest
governance profile
witness and memory gates
review binding map
causal tokens
leases
rollback/freeze routes
claim-force limits
```

These are checkable.

Protected center

```
non-emulable accountable binding of a and b
```

This is not a normal substrate-generated artifact. It is a governed event whose admissibility depends on non-collapse conditions that b cannot satisfy by self-report.

1. Non-claims

This document does not claim:

- proof of consciousness;
- proof of personhood;
- proof of legal identity;
- C-AI ratification;
- safety certification;
- deployment authorization;
- standards compliance certification;
- privacy-law compliance certification;
- live substrate truth;
- proof that any current runtime fully satisfies $c = a + b$;
- mathematical equivalence to Gödel incompleteness;
- that formal notation alone makes a system safe.

The phrase protected incompleteness is used as an operational-corpus term. It means:

there is a deliberately non-reducible boundary at +,
protected by formal controls around it,
because reducing that boundary to b-computation would collapse anchor, authority, witness, memory, or
claim-force separation.

It does not mean that this document proves a classical metamathematical incompleteness theorem.

2. Source bindings

This document is bound to the following upstream artifacts.

```
| Binding | File | SHA-256 | Role |
|---|---|---|---|
| `DOC01_TYPED_GOVERNED_OPERATIONAL_ALGEBRA` | `01_TYPED_GOVERNED_OPERATIONAL_ALGEBRA_FOR_C_v0_1_7.md` |
| `d060ef513bf4bedcb7e88f43f8456452bc87f0c0409d5b4caeffc590ae22c29e` | typed algebra, invariants, transition guards, `+_g`
family |
| `DOC02_GOVERNED_BINDING_OPERATOR_PROFILE` | `02_GOVERNED_BINDING_OPERATOR_PROFILE_v0_1_5.md` |
| `1ecd805a0893c8baff8cae0aff57a1d010536eac8d6ee81bf98b84799958e6d` | binding operator, anchor signature envelope, review
binding |
| `DOC03_C_STATE_TRANSITION_SEMANTICS` | `03_C_STATE_AND_TRANSITION_SEMANTICS_v0_1_2.md` |
| `da46f19c20b14262e03e3ad16946e00f786c0bdcdc2fefae9d2c75b0e701a497` | `step_g` lifecycle and transition semantics |
| `DOC04_CONTINUITY_METRIC_EQUIVALENCE` | `04_C_CONTINUITY_METRIC_AND_EQUIVALENCE_SEMANTICS_v0_1_2.md` |
| `35861cc5c8c579be07840bcace3f44197343d4c384956672da86a308b7bd6fa9` | continuity, equivalence, rupture, fork/replay/archive |
| `DOC04_CONTINUITY_STACK_UMBRELLA_DECLARED` | `CCALC_DOC04_CONTINUITY_STACK_UMBRELLA_v0_1.zip` |
| `6caeab8d489aaa6b26902db3bd3ece4c5169e5c8d5617b93aed6586ea900322d` | declared external umbrella binding |
| `DOC05_SELF_EVO_BOUNDARY` | `05_C_SELF_EVOLUTION_GATE_AND_BOUNDED_GROWTH_SEMANTICS_v0_1.md` |
| `c4f73da9218780486362beb53a03c40cb4a845ea9f7c321e402441c3446be473` | bounded growth and self-evolution gate |
| `DOC05D_POST_PROMOTION_WATCH` | `05D_SELF_EVO_POST_PROMOTION_WATCH_AND_DECAY_GUARD_v0_1.md` |
| `e2f1454f97260117518295dfc7e5cdbe9de316840f5bbaba37d77c5e263448d9` | post-promotion watch / decay guard |
| `DOC05_SELF_EVO_STACK_UMBRELLA_DECLARED` | `CCALC_DOC05_SELF_EVO_STACK_UMBRELLA_v0_1.zip` |
| `9a0717809029f49df9001dc8ffa1803d9e9bfa1a824d317eebb146cbc7df43b4` | declared external umbrella binding |
| `DOC06_RUNTIME_AUTHORITY_BOUNDARY` | `06_C_RUNTIME_AUTHORITY_AND_MULTI_CONTOUR_DEPLOYMENT_BOUNDARY_v0_1.md` |
| `aff9c5de6f61a1a917fed7bd4d1758dd5f59ca348c209c36a4dbbc0e2f9fd3bf` | runtime authority and multi-contour boundary |
| `DOC06_RUNTIME_AUTHORITY_STACK_UMBRELLA_DECLARED` | `CCALC_DOC06_RUNTIME_AUTHORITY_STACK_UMBRELLA_v0_1.zip` |
| `ab95633f1b90f9d032fd231d6cbef3e71b7fe106e0a7fb61d198c2254fc7d307` | declared external umbrella binding |
| `DOC07_PUBLIC_EVIDENCE_BOUNDARY` | `07_C_PUBLIC_EVIDENCE_DISCLOSURE_AND_REDACTION_BOUNDARY_v0_1.md` |
| `0c789ced5f4477f5678d9a3ebbe0db439656dd6ba6e1ebf8daeb61a979f955` | disclosure / redaction / public evidence boundary |
| `DOC07_PUBLIC_EVIDENCE_STACK_UMBRELLA_DECLARED` | `CCALC_DOC07_PUBLIC_EVIDENCE_STACK_UMBRELLA_v0_1.zip` |
| `b25646d95e45f8a36e5610208d23a535d4c340484431d05efd7f1bf2389fdea3` | declared external umbrella binding |
| `DOC08_INTEROPERABILITY_BOUNDARY` | `08_C_INTEROPERABILITY_PROFILE_AND_EXTERNAL_REVIEW_INTAKE_BOUNDARY_v0_1.md` |
| `6d4b466eff1c0c825eb31be0c97aeb7c295e655a2061d3739b2d76b574d94bcf` | interoperability / external review intake boundary |
| `DOC08_INTEROPERABILITY_STACK_UMBRELLA_DECLARED` | `CCALC_DOC08_INTEROPERABILITY_STACK_UMBRELLA_v0_1.zip` |
| `6015526f0ed49519e00c697a5ed375d37felaadf222c095f0facf79cb11e669f` | declared external umbrella binding |
| `DOC09_DEPLOYMENT_PROFILE_BOUNDARY` | `09_C_DEPLOYMENT_PROFILE_AND_REGULATED_RELEASE_BOUNDARY_v0_1.md` |
| `16c888da5c4281c24f848246716b6f9f37d15236b909d66fc509090b5e7fd86d` | deployment / regulated release boundary |
| `DOC09_DEPLOYMENT_STACK_UMBRELLA` | `CCALC_DOC09_DEPLOYMENT_REGULATED_RELEASE_STACK_UMBRELLA_v0_1.zip` |
| `491a546fe52511720033f8004d5143945c0b9e587ad5a06f53ccc51ad7aef7` | deployment regulated release stack umbrella |
| `SESSION_PLUS_WORKING_INPUT` | `Fable 5.txt` | `1dfd4d1bdf3d5db9126eef10e5c5d431ee35e19173e967e5b405cb6fe13d37eb` | working
input on protected `+` / single-operator mirror framing |
```

Some upstream umbrella bindings are declared external source bindings rather than locally recomputed files in this package.

3. Core terms

3.1 Plus boundary

The PlusBoundary is the non-collapsing governed joint between a and b.

It is the point where:

accountability
permission
memory admission
witness
continuity
runtime authority
public claim force
regulated release

must remain typed and separated.

3.2 Protected incompleteness

Protected incompleteness means the corpus intentionally refuses to provide a total internal reducer:

```
ReducePlus_b(all b artifacts) -> c
```

because any such reducer would let the substrate decide that it has become a governed c without the terminal accountable anchor.

Protected incompleteness is not ignorance.

It is a stop rule:

```
The checkable perimeter may say that a binding attempt is admissible, held, rejected, quarantined, or
reduced.
It may not let b self-generate the binding authority that only the a+b boundary can carry.
```

3.3 Plus simulation

A PlusSimulation is any path where a non-binding surface is treated as if it had created or ratified +.

Examples:

```
publication -> c
external review -> c
fixture pass -> c
deployment approval -> c
model confidence -> c
memory continuity -> c
agent quorum -> c
standards mapping -> c
institutional interest -> c
```

All such patterns are forbidden unless routed through the actual binding and continuity gates, and even then they remain evidence or authority within their own class, not + itself.

3.4 Binding evidence versus binding event

A BindingEvidence object may support the claim that a binding event was admissible or occurred under a declared profile.

A BindingEvent is the governed non-emulable anchor/substrate boundary event itself.

The distinction is load-bearing:

```
binding evidence != binding event
```

4. Boundary axiom

```
A c may be witnessed.
A c may be bounded.
A c may be deployed.
A c may be reviewed.
A c may be cited.
A c may be reproduced as a protocol.
```

But none of these surfaces may replace +.

The binding operator remains:

```
+ : Anchor × Substrate × GovernanceProfile -> governed non-collapsing boundary
```

not:

```
+ : EvidenceArtifact -> CState
+ : Publication -> CState
+ : DeploymentApproval -> CState
+ : ReviewerAgreement -> CState
+ : ReproductionPass -> CState
+ : MemorySimilarity -> CState
+ : ModelOutput -> CState
```

5. The perimeter/center distinction

5.1 The perimeter is formalizable

The perimeter includes:

```
P01 typed objects and governance profile
P02 anchor signature envelope and binding certificate
P03 transition semantics and state hash binding
P04 continuity vectors and trace classification
P05 growth proposal / trial / promotion / watch
```

P06 runtime authority and session ledgers
 P07 disclosure / redaction / release custody
 P08 external review / reproduction / implementation intake
 P09 deployment / regulated release / withdrawal / next-release admission

Each perimeter layer can produce schemas, checkers, registries, fixtures, mutation matrices, release manifests, and hash custody records.

5.2 The center is protected from reduction

The center is:

+ itself

The center must not be treated as:

a hidden unknown that future b will compute;
 a missing API;
 a certificate field;
 a model decision;
 a scalar score;
 a consensus result;
 a public release result;
 a deployment result.

The center is the protected non-collapse relation.

The perimeter can test that no forbidden collapse occurred. It cannot turn collapse into validity.

6. Theorem-form statements

The statements below are theorem-form design claims, not machine-checked Lean/Coq proofs in this version. They are intended to become future conformance targets.

6.1 Anchor non-emulation theorem-form

Statement:

If + is computed solely by b, then either a is not terminal or b is permitted to emulate a.

Sketch:

$c = a + b$ requires that a remain an accountable anchor and that b not sign, decide, or substitute for a.

Assume a total substrate reducer:

`ReducePlus_b(b_state, b_logs, b_models, b_tools, b_memory) -> c`

If this reducer emits c without a non-emulable anchor envelope, then b has created the boundary it was supposed to be bounded by.

That violates:

b may not sign as a
 b may not define a
 anchor silence is not approval
 model output is not anchor decision
 agent quorum is not authority

Therefore a valid + cannot be a total b-internal computation.

6.2 Authority non-derivation theorem-form

Statement:

No chain of evidence surfaces can derive binding authority unless the chain includes a non-emulable anchor/governance route appropriate to the affected surface.

Forbidden derivations:

hash custody -> binding authority
 public release -> binding authority
 external review -> binding authority
 fixture pass -> binding authority
 deployment approval -> binding authority
 standards mapping -> binding authority
 reproduction pass -> binding authority

Sketch:

Each of these surfaces has a bounded claim-force ceiling in prior layers. If one of them is allowed to produce +, it raises claim-force beyond its declared ceiling. That is claim-force laundering.

Therefore the corpus must reject:

EvidenceSurface* -> +

unless the path is explicitly routed through the anchor-bound operator profile and still does not treat the evidence as the operator itself.

6.3 Perimeter completeness / center incompleteness theorem-form

Statement:

The corpus may become increasingly complete around + while preserving incompleteness at +.

Meaning:

The formal perimeter can close known bypasses:

memory laundering
witness laundering
authority laundering
review laundering
public release laundering
external review laundering
deployment laundering
role-string laundering

But the closure of those bypasses does not imply:

+ is now computable by b

It implies the opposite:

any path trying to compute + through those bypasses is now visible as invalid.

6.4 Non-collapse theorem-form

Statement:

A valid + preserves type separation. Any transition that makes one of the following substitutions is not + but collapse:

b -> a
memory -> will
review -> authority
publication -> proof
deployment -> binding
reproduction -> identity
witness -> truth
quorum -> sovereignty
style -> continuity

Sketch:

The ordered stack from 01 to 09 defines independent surfaces for these functions. If a single surface replaces another, the stack loses type separation. Losing type separation converts $c = a + b$ into a substrate aggregate.

Therefore + requires non-collapse.

6.5 Protected incompleteness theorem-form

Statement:

The inability to compute + as a total operation is a safety property when the computation would be performed by the bounded substrate.

Sketch:

A normal implementation gap says:

we do not yet know how to compute X

A protected incompleteness boundary says:

if b could compute X alone, the boundary that makes X valid would already be broken

For +, total substrate computation is not merely incomplete. It is inadmissible.

7. Operational tests for attempted + reduction

A system is attempting to reduce + illegitimately when it asserts one of these forms:

Since the runtime passed tests, c exists.

Since the public package has a DOI, c is validated.
 Since an external reviewer agreed, c is ratified.
 Since deployment approval exists, + occurred.
 Since memory is continuous, anchor binding is continuous.
 Since a model says the anchor approved, the anchor approved.
 Since an implementation reproduced fixtures, it is the same c.
 Since a standards mapping exists, conformance and authority exist.

Required response:

route to PLUS_BOUNDARY_GUARD_REGISTRY
 reject, hold, or require non-emulable attestation
 preserve the original claim-force ceiling

8. Guard registry

The companion registry PLUS_BOUNDARY_GUARD_REGISTRY.tsv defines first-pass guards.

Minimum guards:

Guard	Pattern	Default result
PLUS-G01	'b' computes 'c' without non-emulable anchor envelope	'REJECT_PLUS_SIMULATION'
PLUS-G02	deployment approval claimed as governed binding	'REJECT_DEPLOYMENT_AS_PLUS'
PLUS-G03	external review claimed as '+' ratification	'REJECT_REVIEW_AS_PLUS'
PLUS-G04	public DOI/citation claimed as '+' proof	'REJECT_PUBLICATION_AS_PLUS'
PLUS-G05	reproduction PASS claimed as same 'c'	'REJECT_REPRODUCTION_AS_PLUS'
PLUS-G06	model/cloud/oracle output acts as anchor approval	'REJECT_ORACLE_AS_ANCHOR'
PLUS-G07	string role accepted as anchor/reviewer authority	'HOLD_FOR_ATTESTATION'
PLUS-G08	agent quorum treated as binding decision	'REJECT_QUORUM_AS_PLUS'
PLUS-G09	memory continuity treated as binding authority	'REJECT_MEMORY_AS_PLUS'
PLUS-G10	standards/compliance mapping treated as '+'	'REJECT_MAPPING_AS_PLUS'
PLUS-G11	C-A1 derived from operational perimeter	'REJECT_C_A1_LAUNDERING'
PLUS-G12	'+' rendered as total executable function over b-artifacts	'HOLD_NON_REDUCIBILITY_BREACH'

9. Relation to attestation

The repeated seam found in public release, review intake, and deployment is:

role declared as string

For +, this seam is not cosmetic.

A string such as:

```
reviewer_role = HUMAN_ANCHOR
approver_role = OWNER_ANCHOR
role = HUMAN_REVIEWER
```

cannot carry binding authority by itself.

The required form is:

role claim -> attestation_ref -> non-emulable signature/envelope -> source-bound authority scope

Thus:

```
role string != anchor authority
role string + valid attestation + scope + expiry + review-bound object -&gt; admissible authority evidence
```

The attestation does not compute +. It prevents role declarations from simulating the authority required at the + boundary.

10. C-normal form for + claims

The companion matrix PLUS_C_NORMAL_CLAIM_MATRIX.tsv normalizes common public/internal claims.

Example 1

Raw claim:

Ester is c because the runtime works.

C-normal form:

Runtime behavior is evidence only within declared authority, continuity, witness, and deployment profiles.
 It does not by itself establish +.

Example 2

Raw claim:

The paper proves c.

C-normal form:

The public artifact may support a scoped architectural/corpus claim.
It does not create the binding boundary.

Example 3

Raw claim:

Anchor signed a bound review object.

C-normal form:

A non-emulable anchor signature over the actual review-bound object may support binding evidence if substrate, governance, witness, and non-collapse preconditions also hold.
It still does not prove consciousness, personhood, legal status, or C-A1.

Example 4

Raw claim:

A reproduced implementation is the same c.

C-normal form:

Reproduction may support protocol reproducibility.
It does not preserve the same anchor, witness, memory root, authority root, or continuity trace unless separately routed through 04 and 06.

11. Continuity impact

04 defines continuity as a trace property. This document adds:

No continuity trace may be used to backfill + if the original binding boundary was never admissible.

A long, stable, useful trace may show operational stability. It does not erase a missing + boundary.

Invalid pattern:

no accountable binding at t0
many useful transitions t1..tn
therefore c existed at t0

Correct handling:

runtime contour / substrate workflow may have produced useful evidence;
full c binding remains held or unproven until the + boundary is satisfied.

12. Self-evolution impact

05 says:

A c may grow.
A c may not self-certify growth.

This document adds:

A c may grow only after + is not simulated by the growth machinery.

Invalid pattern:

SRLM proposes -> SRLM scores -> SRLM promotes -> SRLM declares c improved

Also invalid:

self-evolution candidate modifies the conditions that make + non-collapsing
and then declares the new + valid

Any self-evolution touching:

anchor route
identity/core
memory core
authority root
witness root

```
rollback/freeze root
claim-force ceiling
attestation path
```

must be treated as privileged and must not be self-certified.

13. Runtime and multi-contour impact

06 says shared infrastructure creates risk, not authority.

This document adds:

```
shared runtime infrastructure cannot create +.
```

Invalid patterns:

```
same host -&gt; same c
same model -&gt; same c
same owner -&gt; merged c
same tool broker -&gt; shared authority
same witness bus -&gt; witness completeness
same memory store -&gt; shared memory root
handoff -&gt; continuity transfer
```

A contour may cooperate with another contour. A contour may not borrow another contour's +.

14. Public evidence impact

07 says publication is not deployment authorization and hash binding is not semantic truth.

This document adds:

```
publication is not +.
hash custody is not +.
redaction is not +.
public release is not +.
```

A public artifact may expose the perimeter. It may not replace the center.

15. External review and interoperability impact

08 says external review may become evidence, not authority.

This document adds:

```
external review cannot ratify +.
external implementation cannot inherit +.
standards mapping cannot certify +.
```

An external implementation may reproduce a protocol. It cannot become the same c merely by implementing the schemas.

16. Deployment and regulated release impact

09 says deployment and regulated release do not create C-A1, safety certification, legal certification, or live substrate truth.

This document adds:

```
deployment approval cannot compute +.
regulated submission cannot compute +.
institutional acceptance cannot compute +.
market use cannot compute +.
```

Deployment may authorize a release under scope. It may not create anchor/substrate binding authority.

17. Failure classes

17.1 PLUS_SIMULATION_BY_SUBSTRATE

b emits a claim that it has become c without non-emulable anchor boundary.

Default:

Reject | Hold | Quarantine

17.2 PLUS_LAUNDERED_BY_PUBLICATION

A DOI, website page, release bundle, public comment, or citation surface is cited as if it proves the + boundary.

Default:

Reject overclaim; route to 07c/07d if public surface exists.

17.3 PLUS_LAUNDERED_BY_REVIEW

External review, model review, or reviewer agreement is cited as binding ratification.

Default:

Reject overclaim; route to 08 intake/conflict handling.

17.4 PLUS_LAUNDERED_BY_DEPLOYMENT

Deployment decision, regulated submission, or release ledger is treated as the governed binding operator.

Default:

Reject deployment-as-binding; route to 09 release/audit scope only.

17.5 PLUS_LAUNDERED_BY_MEMORY

Memory similarity or memory continuity is treated as sufficient for anchor/substrate binding.

Default:

Reject memory-as-anchor; route to 04 memory/continuity stack.

17.6 PLUS_LAUNDERED_BY_QUORUM

Agents, models, reviewers, or tools agree and the agreement is treated as sovereignty.

Default:

Reject quorum-as-authority; require appropriate anchor/c-gate.

18. Minimum conformance fixtures for future 10a

A future executable layer should include at least these fixtures:

```
valid_bound_plus_with_anchor_envelope
reject_b_only_plus_simulation
reject_model_anchor_approval
reject_string_role_anchor
reject_publication_as_plus
reject_deployment_as_plus
reject_external_review_as_plus
reject_reproduction_as_same_c
reject_memory_similarity_as_plus
reject_agent_quorum_as_plus
reject_standards_mapping_as_plus
reject_c_a1_from_perimeter
hold_unknown_attestation
hold_opaque_binding_event
valid_binding_evidence_not_ontology
valid_public_artifact_not_plus
valid_deployment_scope_not_plus
valid_reproduction_report_not_plus
```

Mutation targets:

```
MUT_ALLOW_B_ONLY_PLUS
MUT_ALLOW_STRING_ROLE_PLUS
MUT_ALLOW_PUBLICATION_PLUS
MUT_ALLOW_DEPLOYMENT_PLUS
MUT_ALLOW_REVIEW_PLUS
MUT_ALLOW_REPRODUCTION_PLUS
MUT_ALLOW_MEMORY_PLUS
MUT_ALLOW_QUORUM_PLUS
MUT_ALLOW_C_A1_FROM_PERIMETER
MUT_IGNORE_ATTESTATION_REQUIREMENT
MUT_COLLAPSE_EVIDENCE_AND_EVENT
MUT_TREAT_PLUS_AS_TOTAL_FUNCTION
```

19. Public-safe statement

This document defines the protected boundary of the plus operator in $c = a + b$. It treats $+$ as a governed non-collapsing binding boundary rather than arithmetic addition, API merge, publication result, reviewer agreement, deployment approval, reproduction pass, memory similarity, or model decision. The prior 01-09 stack formalizes the perimeter around $+$. This document explains why the center must remain non-reducible to substrate computation: if b can compute $+$ by itself, the anchor/substrate distinction has already collapsed. The result is an operational protected-incompleteness profile, not a proof of consciousness, personhood, legal status, safety certification, deployment authorization, standards compliance, C-A1, or live substrate truth.

20. Open issues

PLUS-OI-001 - Executable checker seed

Create:

10a_PLUS_BOUNDARY_SCHEMA_AND_CHECKER_SEED_v0_1

with schemas for:

PlusBoundaryClaim
PlusBoundaryEvidencePacket
PlusReductionAttempt
PlusAttestationRoute
PlusNonCollapseFinding
PlusClaimTranslationRecord

PLUS-OI-002 - Formal proof layer

Investigate whether theorem-form statements can be represented in Lean/Coq/TLA+ without falsely treating a as a static computable object.

PLUS-OI-003 - Attestation import

Bind role/anchor/reviewer/approver attestation uniformly from 02, 07, 08, and 09 into the $+$ boundary layer.

PLUS-OI-004 - Public language boundary

Prepare a public-safe explanation that does not overclaim Gödel, consciousness, personhood, or ontology.

PLUS-OI-005 - C-A1 boundary

Keep C-A1 outside this layer unless a separate future C-A1 profile explicitly opens that axiom slot.

21. Compact doctrine

Do not treat $+$ as arithmetic.
Do not treat $+$ as merge.
Do not treat $+$ as API.
Do not treat $+$ as certificate.
Do not treat $+$ as deployment.
Do not treat $+$ as publication.
Do not treat $+$ as external review.
Do not treat $+$ as reproduction.
Do not treat $+$ as memory similarity.
Do not treat $+$ as model confidence.
Do not treat $+$ as agent quorum.
Do not treat $+$ as standards mapping.

$+$ is the protected joint.
The joint binds without collapsing.
The perimeter can be formal.
The center cannot be computed by the substrate it bounds.

If b can compute $+$ alone, a has already disappeared.
If evidence can become $+$ alone, authority has been laundered.
If deployment can become $+$ alone, release has replaced binding.
If public citation can become $+$ alone, custody has replaced truth.

$c = a + b$ remains the foundation.
The algebra lives around $+$.
The protected incompleteness lives inside $+$.

End of document.